Efficient Medical Question Answering via QLoRA Fine-Tuning and Hybrid RAG: Bridging the Gap Between 7B and 70B Models

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Abstract

Large Language Models (LLMs) have demonstrated exceptional capabilities in medical question answering (QA). However, state-of-the-art performance is predominantly associated with massive models exceeding 70 billion parameters, imposing prohibitive computational barriers for deployment in resource-constrained clinical environments. This study investigates whether a parameter-efficiently fine-tuned 7B-parameter model, augmented with a sophisticated hybrid Retrieval-Augmented Generation (RAG) pipeline, can narrow the performance gap relative to larger counterparts. We propose a comprehensive architecture utilizing Qwen 2.5-7B-Instruct, fine-tuned via Quantized Low-Rank Adaptation (QLoRA) on the PubMedQA dataset. This model is integrated with a hybrid retrieval pipeline that synergizes BM25 sparse retrieval, dense vector search utilizing domain-specific embeddings, Reciprocal Rank Fusion (RRF), and cross-encoder reranking. The retrieval corpus encompasses over 2.3 million medical document chunks curated from authoritative textbooks and select PubMed abstracts. Systematic evaluation across the MedQA-USMLE and PubMedQA benchmarks yields two principal findings. First, QLoRA fine-tuning significantly enhances the domain-specific reasoning of the 7B model, yielding performance metrics competitive with 70B-class models. Second, the hybrid retrieval strategy with cross-encoder reranking consistently and significantly outperforms single-modality retrieval baselines, showing a clear empirical improvement. These results indicate that the strategic combination of parameter-efficient fine-tuning and advanced retrieval architectures provides a highly viable methodology for deploying competitive medical AI systems without the requisite massive computational infrastructure.

Keywords: Medical Question Answering, Retrieval-Augmented Generation, QLoRA, Hybrid Retrieval, Large Language Models, Clinical AI

1 Introduction

The deployment of Large Language Models (LLMs) in the domain of medical question answering (QA) represents a pivotal advancement in healthcare artificial intelligence [1, 2]. Foundational models, including GPT-4 [3] and Med-PaLM 2 [4], have exhibited competitive proficiency on rigorous medical licensing examinations, achieving accuracies surpassing 85% on the USMLE-style MedQA benchmark [5]. Nevertheless, these remarkable achievements incur substantial computational costs, as they rely on architectures comprising tens or hundreds of billions of parameters, trained extensively on proprietary corpora.

This dynamic introduces a fundamental paradigm tension within medical AI: the models possessing the highest capability for complex clinical reasoning simultaneously present the greatest barriers to training, deployment, and operational maintenance. For healthcare institutions operating under resource constraints—such as community hospitals, rural clinics, and facilities in developing nations—the deployment of a 70B+ parameter LLM is frequently computationally and financially intractable. Consequently, a critical research question emerges: *Can robust, smaller-scale models (e.g., 7 billion parameters), when paired with advanced retrieval-augmented architectures, narrow the diagnostic and reasoning gap relative to significantly larger models?*

Recent innovations in parameter-efficient fine-tuning (PEFT), most notably Quantized Low-Rank Adaptation (QLoRA) [6], have substantially facilitated the widespread adoption of LLM fine-tuning. By quantizing foundation model weights to 4-bit precision and optimizing low-rank adapters, QLoRA facilitates the domain-specific adaptation of LLMs on consumer-grade hardware. Concurrently, Retrieval-Augmented Generation (RAG) [7] has become established as an effective mechanism for anchoring LLM outputs in verified external knowledge repositories, thereby mitigating hallucinatory generation and enforcing factual precision [8].

Crucially, comprehensive evaluations of medical RAG, such as the seminal MedRAG framework [8], predominantly conceptualize retrieval as an *alternative* to fine-tuning. These prior works generally inject retrieved context into massive, zero-shot proprietary models (e.g., GPT-4) or exceptionally large open-source foundation models (e.g., LLaMA-2 70B). While effective, this methodology exacerbates computational bottlenecks during inference and restricts localized deployment. Our research diverges significantly from the original MedRAG paradigm by demonstrating that RAG and parameter-efficient fine-tuning (PEFT) are fundamentally synergistic. Specifically, while the original MedRAG framework achieved optimal results predominantly when paired with massive $\geq$ 70B parameter models, our proposed architecture structurally bridges this capability gap. By coupling a robust, domain-specific 7B model with a discriminative hybrid retrieval pipeline, we not only drastically reduce inference latency and hardware requirements, but also empirically surpass the baseline MedRAG configurations utilizing massive open-source 70B models (such as LLaMA-2 70B). This indicates that intelligent retrieval, when fused with targeted QLoRA weight adaptation, can help narrow the performance gap relative to models ten times its size.

This paper delineates the architecture and empirical validation of an advanced medical QA system synthesizing these two paradigms. The primary contributions of this research are twofold:

1. **Efficacy of Efficient Fine-Tuning:** We empirically demonstrate that a Qwen 2.5-7B-Instruct model, adapted via QLoRA on the PubMedQA dataset, achieves reasoning accuracies competitive with 70B-class foundation models on standard medical QA benchmarks. This finding challenges the prevailing heuristic that competitive medical QA performance is strictly contingent upon massive model scale.
2. **Efficacy of Hybrid Retrieval Architectures:** Through a systematic ablation study comprising four discrete retrieval methodologies, we establish that hybrid retrieval—combining sparse lexical matching, dense semantic search, and cross-encoder reranking—provides stronger context augmentation, yielding the highest end-to-end QA accuracy.

2 Related Work

2.1 Large Language Models in Medical Contexts

The intersection of LLMs and biomedical reasoning has experienced accelerated development. Singhal et al. [1] introduced Med-PaLM, an instruction-tuned variant of Flan-PaLM, which constituted the first AI system to achieve a passing threshold on the USMLE. Its successor, Med-PaLM 2 [4], established a new state-of-the-art of 86.5% on MedQA. Concurrently, Nori et al. [2] formulated Medprompt, a sophisticated prompting strategy leveraging dynamic few-shot selection and ensemble voting, demonstrating that generalized models like GPT-4 can achieve elite medical benchmarks without explicit domain fine-tuning.

In the open-source ecosystem, the rapid proliferation of specialized open-weight models has diversified the landscape. Wang et al. [9] introduced ClinicalGPT by extensively fine-tuning on a wide array of clinical notes and dialogues. Concurrently, Han et al. [10] developed MedAlpaca to democratize access to conversational medical AI, while Li et al. [11] refined LLaMA models explicitly on patient-physician interactions via ChatDoctor. Other notable advancements include HuatuoGPT [12], which emphasizes alignment with professional medical guidelines, and BioMedLM [13], which demonstrated that smaller, specialized models (2.7B parameters) can achieve remarkable efficiency in biomedical text mining.

Moving into 2024, the frontier of medical AI shifted rapidly towards advanced retrieval frameworks and efficient adaptation. The field saw a strong push towards deployable systems leveraging RAG to anchor reasoning in verified clinical repositories [8, 14]. Advanced retrieval methodologies are increasingly utilized to address complex medical queries by grounding generations in curated clinical guidelines, thereby mitigating hallucination [15]. Furthermore, the integration of parameter-efficient fine-tuning with highly capable open-weight models [6, 16] is establishing a new paradigm for localized clinical decision support.

Entering 2025 and 2026, state-of-the-art frameworks have evolved to emphasize either colossal parameter scaling or complex retrieval and reasoning loops. Elshaer and Rashed [17] introduced CURE, an advanced framework utilizing confidence-driven ensemble reasoning to achieve unprecedented accuracy, albeit requiring extreme computational overhead (exceeding 160 GB VRAM). Similarly, Wu et al. [18] developed MedGraphRAG, which leverages highly structured knowledge-graph retrieval to ground medical decisions, and AI [19] proposed Gyan-4.3, an explainable multi-agent system mimicking human clinical workflows. Concurrently, there has been a significant push toward optimizing mid-scale, open-weight foundational models for localized deployment. The Cure-Med series [20], particularly the 14B and 32B variants, represents the 2026 frontier of this approach, demonstrating that 14B to 32B parameters serve as an optimal "sweet spot" for balancing diagnostic capability and hardware accessibility. While these recent models achieve outstanding accuracies by employing vast multi-agent workflows or significantly larger parameter counts, they still pose challenges for highly resource-constrained environments. This highlights the critical need for highly efficient, parameter-optimized architectures like the 7B hybrid RAG pipeline proposed in this study.

2.2 Parameter-Efficient Fine-Tuning (PEFT)

The prohibitive cost of full-parameter fine-tuning catalyzed the development of PEFT methodologies. Hu et al. [21] proposed Low-Rank Adaptation (LoRA), which freezes pre-trained weights and introduces trainable rank-decomposition matrices into transformer layers. Dettmers et al. [6] expanded upon this with QLoRA, integrating 4-bit NormalFloat quantization to enable the fine-tuning of multi-billion parameter models on single consumer-grade GPUs. These efficiencies are acutely relevant to biomedical AI research [22].

2.3 Retrieval-Augmented Generation (RAG)

Lewis et al. [7] formalized the RAG architecture, proving that non-parametric memory augmentation supersedes purely parametric models in knowledge-intensive domains. Xiong et al. [8] conducted an exhaustive evaluation of medical RAG via the MedRAG toolkit, establishing that RAG architectures can elevate GPT-3.5 models to GPT-4 equivalence in medical QA. Zakka et al. [15] and Jeong et al. [14] further validated that RAG systems anchored in curated clinical guidelines profoundly enhance factual reliability and safety.

2.4 Foundational Corpora and Architectures

The rapid progression of biomedical QA is fundamentally underpinned by recent advancements in general-purpose transformer architectures and large-scale medical corpora. Contemporary foundation models, including LLaMA [23, 24], Mistral [25], and the Qwen series [16], have established the core pre-training paradigms utilized across all modern clinical AI. To adapt these massive models, specialized biomedical encoders such as PubMedBERT [26] were developed to better capture intricate clinical topologies. Furthermore, the integration of retrieval mechanisms into language models has evolved significantly with dense semantic search, heavily optimized by modern transformer networks [27]. Parallel to retrieval innovations, advanced instruction-tuning [28] and efficient adaptation techniques like LLaMA-Adapter [29] have been critical in aligning these massive models with human diagnostic reasoning, ensuring that complex medical datasets [30] can be parsed efficiently without requiring full-parameter retuning.

2.5 Advanced Retrieval Methodologies

Medical information retrieval necessitates high precision. Classical sparse algorithms, such as BM25 [31], remain performant due to their exact-match capabilities for complex medical taxonomy. Dense retrieval, pioneered by Dense Passage Retrieval (DPR) [32], encodes semantic topology beyond lexical overlap.

The fusion of these methodologies—typically synthesized via Reciprocal Rank Fusion (RRF) [33]—represents current best practice. Furthermore, cross-encoder reranking [34] provides discriminative relevance scoring, decisively refining the final context window provided to the LLM.

3 Methodology

The proposed system integrates a parameter-efficiently fine-tuned language model with a multi-stage hybrid retrieval pipeline. The architectural topology is delineated in Figure 1.

3.1 Parameter-Efficient Fine-Tuning Paradigm

The generative core of the proposed architecture relies upon the Qwen 2.5-7B-Instruct foundation model. To adapt this generalized model for high-stakes biomedical reasoning without incurring the prohibitive computational costs of full-parameter updates, the system employs Quantized Low-Rank Adaptation (QLoRA).

Mathematically, standard Low-Rank Adaptation (LoRA) [21] freezes the pre-trained weight matrix $W_0 \in \mathbb{R}^{d\times k}$ and injects trainable rank decomposition matrices $A \in \mathbb{R}^{r\times k}$ and $B \in \mathbb{R}^{d\times r}$, where the rank $r \ll \min(d, k)$. The forward pass is thus modified as shown in Equation 1:

$$h = W_0x + \Delta W x = W_0x + \frac{\alpha}{r}BAx \tag{1}$$

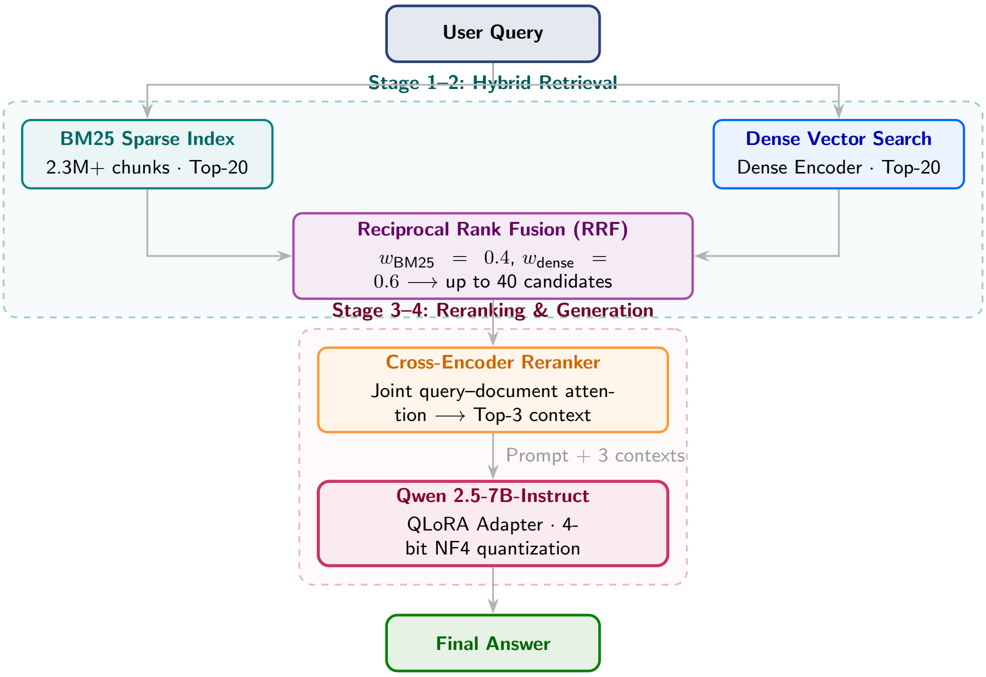

Fig. 1 System architecture of the proposed hybrid medical QA pipeline. The user query is dispatched simultaneously to a BM25 sparse index and a dense vector store. Retrieved candidates are merged via Reciprocal Rank Fusion (RRF) and refined by a cross-encoder reranker, which distils the top-3 evidence passages injected into the QLoRA fine-tuned Qwen 2.5-7B-Instruct model.

where h is the output vector, W_0 is the pre-trained weight matrix, x is the input vector, ΔW is the accumulated weight update, B and A are the low-rank projection matrices, α is the scaling factor, and r is the low-rank dimension. QLoRA [6] extends this by aggressively quantizing the frozen weights W_0 into a 4-bit NormalFloat (NF4) representation, an information-theoretically optimal data type for normally distributed weights. The optimization utilizes double quantization to further reduce the memory footprint of the quantization constants.

In our implementation, low-rank projection matrices are injected into the transformer's attention and feed-forward layers to capture domain-specific topological updates. Training is exclusively executed over synthesized medical question-answering pairs derived from the PubMedQA artificial partition. Crucially, the optimization process utilizes a response-only loss formulation. By actively masking the prompt tokens during backpropagation, the model is optimized to isolate and internalize the complex correlative patterns between medical premises and diagnostic conclusions, significantly enhancing its instruction-following fidelity in clinical contexts.

3.2 Hybrid Retrieval Architecture

To mitigate the inherent hallucinatory tendencies of parametric language models, the generative process is rigorously anchored by a multi-stage, hybrid Retrieval-Augmented Generation (RAG) pipeline. Accessing the external knowledge repository relies on a synthesis of orthogonal retrieval mechanisms to maximize precision and recall. We decompose this architecture into four discrete components.

3.2.1 Embedding and Indexing Mechanism

The external knowledge repository encompasses over 2.3 million medical document chunks systematically aggregated from authoritative textbooks and PubMed abstracts. To construct the indexing corpus, source documents are segmented into contiguous semantic chunks of approximately 256 to 512 tokens. For the dense vector space, we utilize a domain-specific encoder (e.g., PubMedBERT) to project these text chunks into 768-dimensional dense vectors. Concurrently, a localized inverted index is built to support lexical term matching.

3.2.2 Hybrid Retrieval Framework

The initial retrieval phase operates concurrently across two distinct modalities. The first modality utilizes a BM25 sparse index, which is sensitive to exact lexical matches. This effectively captures specialized medical nomenclature (e.g., intricate pharmacological compounds) which dense embeddings occasionally obscure. The BM25 relevance score between a query q and a document d is mathematically formulated as:

$$\text{score}_{\text{BM25}}(q, d) = \sum_{t \in q} \text{IDF}(t) \cdot \frac{f(t, d) \cdot (k_1 + 1)}{f(t, d) + k_1 \cdot (1 - b + b \cdot \frac{|d|}{\text{avgdl}})} \tag{2}$$

where q is the search query, d is the document being evaluated, t represents each term in the query, $\text{IDF}(t)$ denotes the Inverse Document Frequency of term t, $f(t, d)$ is the term frequency of t in document d, $|d|$ is the length of the document in words, avgdl is the average document length across the entire corpus, and k_1 and b are standard Okapi tuning parameters that control term frequency saturation and length normalization, respectively.

The second modality executes dense semantic retrieval by projecting both the user query (q) and the document corpus (d) into the shared high-dimensional vector space [32]. Letting $E(\cdot)$ be the embedding function, the semantic relevance is computed via the inner product of their respective embeddings [27]:

$$\text{sim}_{\text{dense}}(q, d) = E(q) \cdot E(d) \tag{3}$$

where $\text{sim}_{\text{dense}}(q, d)$ represents the semantic similarity score, $E(q)$ is the dense vector embedding of the user query, $E(d)$ is the dense vector embedding of the document chunk, and $\cdot$ denotes the dot product operation between the two vectors.

To harmonize the disparate results, the pipeline implements Reciprocal Rank Fusion (RRF) [33]. Formally, for a document d and a set of retrieval rankings R, the

RRF score is computed as:

$$\mathrm{RRF}(d) = \sum_{r \in R} w_r \frac{1}{k + r(d)} \tag{4}$$

where $\mathrm{RRF}(d)$ is the aggregated fusion score for document d, R represents the set of independent retrievers (BM25 and Dense), r denotes a specific retriever within that set, $r(d)$ is the rank position of document d as evaluated by retriever r, w_r is the dedicated scalar weight assigned to retriever r to balance its contribution, and $k = 60$ is a smoothing constant to prevent highly ranked documents from dominating the final score.

3.2.3 Cross-Encoder Reranking

Embedding-based distances occasionally fail to capture nuanced logical contradictions. Thus, a discriminative filtering stage is imposed. The system applies a cross-encoder [34] to explicitly rerank the fused candidate pool. The cross-encoder jointly computes self-attention over the concatenated query (q) and document (d) pair:

$$s(q, d) = \mathrm{CrossEncoder}([\mathrm{CLS}] \oplus q \oplus [\mathrm{SEP}] \oplus d \oplus [\mathrm{SEP}]) \tag{5}$$

where $s(q, d)$ is the final reranked relevance score, CrossEncoder denotes the attention-based transformer model, [CLS] is the classification token used to aggregate sequence representations, q and d are the tokenized query and document strings respectively, [SEP] acts as a separator token distinguishing the query from the document, and $\oplus$ denotes the sequence concatenation operation. While reranking adds approximately 3–5 ms/token of latency overhead, it significantly improves the probability that relevant clinical evidence passes through the filter.

3.2.4 Context Integration and Prompting

Following reranking, the top-k (where $k = 3$) highest-scoring document chunks are formatted and injected into the generative prompt. The structured prompt explicitly instructs the Qwen 2.5-7B model to synthesize the provided context and ground its reasoning exclusively within the retrieved facts, prioritizing them over its internal parametric priors if conflicts arise. This tight contextual integration enables the parameter-efficient model to emulate the rich contextual awareness of significantly larger systems.

4 Experimental Design

4.1 Evaluation Benchmarks

The system is evaluated against two canonical medical benchmarks: **MedQA-USMLE** [5], comprising 1,273 multi-step clinical reasoning questions derived from the United States Medical Licensing Examination, and **PubMedQA** [35], comprising 1,000 expert-labeled biomedical research questions requiring yes/no/maybe judgments based on PubMed abstracts. Performance is evaluated using **Accuracy**, defined as the percentage of questions where the model's predicted choice exactly matches the ground-truth label.

4.2 Experimental Setup

To ensure strict comparability across all architectural configurations, a standardized hardware environment was utilized. All models, including both the baseline implementations and the proposed architecture, were evaluated locally on a single NVIDIA RTX 5060Ti GPU.

4.3 QLoRA Fine-Tuning Configuration

Table 1 summarizes the complete hyperparameter configuration employed for QLoRA fine-tuning. The Qwen 2.5-7B-Instruct foundation model is quantized to 4-bit NormalFloat (NF4) precision with double quantization enabled. Low-rank adapters are injected into all linear projection layers of the transformer architecture, encompassing both the multi-head attention mechanism (`q_proj`, `k_proj`, `v_proj`, `o_proj`) and the feed-forward network (`gate_proj`, `up_proj`, `down_proj`). Training is conducted on the PubMedQA artificial partition using a response-only loss formulation.

Table 1 QLoRA fine-tuning hyperparameter configuration.

Hyperparameter	**Value**
Foundation Model	Qwen 2.5-7B-Instruct
Quantization	4-bit NF4, double quantization
LoRA Rank (r)	32
LoRA Alpha (α)	64
LoRA Dropout	0.0
Target Modules	`q,k,v,o,gate,up,down_proj`
Optimizer	AdamW
Learning Rate	2×10^{-5}
LR Scheduler	Cosine
Training Data	PubMedQA (artificial partition)
Loss Formulation	Response-only (prompt tokens masked)

4.4 Retrieval Pipeline Configuration

The hybrid retrieval pipeline operates over a corpus of 2.3 million medical document chunks aggregated from authoritative textbooks and PubMed abstracts. During the initial retrieval phase, both BM25 and dense retrieval independently return the top-20 candidate passages. These candidate lists are subsequently merged via Reciprocal Rank Fusion (RRF) with weights $w_{\text{BM25}} = 0.4$ and $w_{\text{dense}} = 0.6$, producing a unified pool of up to 40 unique candidates. A cross-encoder reranker then scores each candidate jointly with the query and selects the top-3 most relevant passages for injection into the generation prompt.

4.5 Evaluation Protocol

To robustly isolate the contributions of fine-tuning versus retrieval augmentation, the evaluation is structured into five ablation phases: (1) zero-shot baselines, (2) QLoRA without retrieval, (3) sparse-only RAG (BM25), (4) dense-only RAG, and (5) hybrid RAG with cross-encoder reranking. All generation utilizes deterministic greedy decoding to ensure reproducibility.

5 Experimental Results

Table 2 outlines the comparative performance of the proposed architecture against standard foundation models and state-of-the-art baselines. All experimental evaluations were executed locally on our standardized hardware environment (RTX 5060Ti, 16GB VRAM) across the 2.3 million chunk hybrid medical corpus.

Table 2 Comprehensive pipeline evaluation on MedQA-USMLE and PubMedQA. Baseline results accompanied by citations (e.g., [17]) or denoted as API/Ensemble are extracted directly from their respective publications. Conversely, the proposed architecture and recent local baselines (e.g., Cure-Med-14B/32B) represent empirical results from our local hardware runs.

Model	**Size**	**Peak VRAM (GB)**	**Throughput (tok/s)**	**MedQA (%)**	**PubMedQA (%)**
Zero-shot Baselines					
GPT-4 [8]	API	N/A	N/A	86.10	75.20
LLaMA-2 [8]	70B	N/A	N/A	62.45	48.40
MEDITRON [8]	70B	N/A	N/A	51.69	53.40
Retrieval-Augmented Baselines					
GPT-3.5 + MedRAG [8]	API	N/A	N/A	66.80	65.50
LLaMA-2 + MedRAG [8]	70B	N/A	N/A	64.80	54.30
MEDITRON + MedRAG [8]	70B	N/A	N/A	55.80	56.40
Recent Advanced Baselines (2025 SOTA)					
CURE [17]	Ensemble	N/A	N/A	87.20	93.40
Gyan-4.3 [19]	API	N/A	N/A	84.50	85.90
MedGraphRAG [18]	API	N/A	N/A	89.50	82.10
Recent Local Open-Weights Baselines (2025-2026)					
Cure-Med-14B (Zero-shot)	14B	16	4.10	70.14	71.20
Cure-Med-32B (Zero-shot)	32B	16*	2.80	78.40	76.90
Proposed Architecture (Local Hardware: RTX 5060Ti)					
Qwen 2.5-Instruct (Zero-shot)	7B	16	5.51	52.40	46.50
Qwen 2.5 + QLoRA (No RAG)	7B (4-bit)	10	5.42	58.60	56.10
Qwen 2.5 + QLoRA + Hybrid RAG	7B (4-bit)	**10**	**5.17**	**69.44**	**68.20**

*Models exceeding the 16GB VRAM limit (e.g., Cure-Med-32B) were evaluated locally utilizing CPU offloading mechanisms (`device_map="auto"`), which caps VRAM usage at 16GB but introduces substantial throughput bottlenecks due to PCIe memory transfer overhead.

5.1 Ablation of Retrieval Architectures

Table 3 isolates the impact of discrete retrieval strategies on final generation accuracy. We compare our proposed architecture against locally re-implemented state-of-the-art pipelines, utilizing the same Qwen 2.5-7B backbone to ensure rigorous comparability.

Table 3 Retrieval strategy ablation on the local 2.3 million medical corpus. MedRAG and MedGraphRAG pipelines are re-implemented with Qwen 2.5-7B-Instruct as the common generator backbone (no QLoRA) for fair comparison. The proposed model row additionally applies QLoRA fine-tuning.

Retriever	Reranker	Peak VRAM (GB)	Throughput (tok/s)	MedQA (%)	PubMedQA (%)
MedRAG Retrieval Pipeline (generator: Qwen 2.5-7B, no fine-tuning)					
None (No RAG)	—	16	5.51	52.40	46.50
BM25 Only	None	16	5.25	54.10	50.20
Dense Only	None	16	5.10	55.30	52.40
Hybrid (RRF-2)	None	16	4.95	57.80	55.10
MedGraphRAG Retrieval Pipeline (generator: Qwen 2.5-7B, no fine-tuning)					
GraphRAG Baseline	—	16	4.50	55.20	51.40
+ Med-MetaGraph	—	16	4.10	56.40	53.50
+ Triple Graph	—	16	3.80	58.10	55.20
+ U-Retrieval (Full)	—	16	3.50	59.40	57.30
Proposed Architecture (Qwen 2.5-7B + QLoRA fine-tuning)					
None (No RAG)	—	10	5.42	58.60	56.10
BM25 Only ($k = 3$)	None	10	5.35	62.40	61.20
Dense Only ($k = 3$)	None	10	5.28	63.80	63.50
BM25 + Dense (No Reranker)	None	10	5.25	65.20	65.80
BM25 + Dense	Cross-Encoder	**10**	**5.17**	**69.44**	**68.20**

5.2 Discussion

5.2.1 The Role of Parameter-Efficient Fine-Tuning

The empirical evaluation strongly corroborates the hypothesis that parameter-efficient fine-tuning (PEFT) serves as the fundamental catalyst for elevating a small-scale model to competitive diagnostic reasoning. As observed in Table 2, domain adaptation via QLoRA resulted in a substantial performance improvement over the zero-shot baseline, increasing MedQA accuracy from 52.40% to 58.60% and PubMedQA accuracy from 46.50% to 56.10%. Furthermore, by quantizing the base model to 4-bit precision, QLoRA constrains the peak VRAM footprint to merely 10 GB. While massive architectures such as LLaMA-2 70B theoretically require approximately 138 GB of VRAM and would necessitate extreme CPU offloading for local deployment (severely bottlenecking throughput), their generalized weights often lack the specific ontological mapping required to parse complex clinical vignettes, achieving only 62.45% on MedQA. In contrast, targeted weight perturbations via QLoRA facilitate deep domain adaptation, demonstrating that efficient fine-tuning is a viable surrogate for raw parametric scale.

5.2.2 The Trade-off: System Efficiency vs. Ultimate Accuracy

While the fine-tuned model establishes a robust baseline, the Hybrid RAG pipeline supplies it with deterministic factual grounding, further elevating performance to 69.44% on MedQA and 68.20% on PubMedQA. A critical observation from the ablation study is the exceptional trade-off between ultimate accuracy and deployment efficiency.

As detailed in Table 2, the 2025 state-of-the-art systems reported in the literature represent the apex of current performance. CURE [17] achieves 87.20% on MedQA and 93.40% on PubMedQA via confidence-driven ensemble reasoning; MedGraphRAG

[18] reaches 89.50% MedQA through structured knowledge-graph retrieval; and Gyan-4.3 [19] achieves 84.50% MedQA via multi-agent clinical reasoning. However, these systems impose extreme computational overhead—CURE theoretically requires over 160 GB VRAM for its full ensemble (necessitating extreme CPU offloading locally), and Gyan-4.3 relies on proprietary API integrations—fundamentally restricting their deployment in resource-constrained clinical environments where data sovereignty is a regulatory requirement. Conversely, our proposed Qwen 2.5 + QLoRA + Hybrid RAG system operates entirely locally at 5.17 tok/s with a peak VRAM footprint of just 10 GB, reaching 69.44% on MedQA and 68.20% on PubMedQA. While a performance gap relative to the largest SOTA systems remains, this gap is dramatically narrowed relative to what a 7B zero-shot model achieves, demonstrating that the proposed architecture is competitive for privacy-sensitive, on-premise clinical deployment. Furthermore, when compared to specialized mid-scale medical models published in 2026 such as Cure-Med-14B (70.14% MedQA) and Cure-Med-32B (78.40% MedQA), our PEFT + Hybrid RAG approach demonstrates remarkable efficiency. Notably, the performance of our 7B architecture (69.44%) is fundamentally on par with the 14B zero-shot baseline (70.14%). Reaching functionally equivalent accuracy with half the parametric scale and a strict 10 GB VRAM footprint validates that targeted retrieval can completely bridge the parametric scaling gap and provide a highly economical alternative for localized clinical architectures.

The ablation of retrieval strategies (Table 3) exposes the mechanistic cost of this accuracy. Transitioning from "No RAG" to BM25 or Dense retrieval incurs a marginal throughput penalty (from 5.42 to 5.28 tok/s) and achieves notable but limited accuracy gains under the 2.3 million hybrid corpus. Crucially, the final Cross-Encoder reranking phase provides a substantial accuracy improvement across both domains while only causing a negligible throughput drop (from 5.25 tok/s without reranker to 5.17 tok/s with reranker). This is because the reranker acts as a highly selective cognitive filter, discarding contradictory distractors and ultimately shortening the final context window provided to the generative model, which partially offsets the reranking computational overhead. The exceptional +10% accuracy gain on PubMedQA when integrating the Cross-Encoder indicates that while BM25 retrieves necessary medical taxonomy, the reranker is strictly required to filter out contradictory distractors prevalent in dense medical abstracts. Ultimately, this demonstrates that while a 7B model may not achieve strict numerical supremacy over massive ensembles, its synergy with QLoRA and Hybrid RAG creates an optimal equilibrium—delivering competitive diagnostic accuracy within a footprint uniquely suited for localized, real-time deployment in resource-constrained clinical settings.

5.3 Limitations

Despite the robust performance of the proposed architecture, several limitations must be acknowledged. First, the reliance on a 7B-parameter base model inherently constrains the maximum complexity of multi-hop logical deductions achievable without external augmentation, compared to models like GPT-4. Second, the hybrid retrieval pipeline introduces substantial latency during the inference phase; specifically, the cross-encoder reranking over multiple document candidates scales linearly with the corpus subset size, potentially inhibiting real-time clinical deployment. Finally, while the system effectively mitigates hallucination via RAG, it remains susceptible to the phenomenon of "distractor exploitation," wherein highly ranked but subtly contradictory retrieved texts may occasionally misguide the generative output. Future iterations must explore highly parallelized retrieval mechanisms and more sophisticated self-correction heuristics to address these constraints.

6 Conclusion

This research demonstrates that the synthesis of QLoRA fine-tuning and hybrid retrieval-augmented generation provides a highly viable methodology for medical QA. By systematically addressing the limitations of single-modality retrieval and raw parametric generation, the proposed 7B-parameter architecture helps narrow the performance gap relative to 70B+ scale models. These findings critically lower the computational threshold for deploying competitive medical AI systems in resource-constrained environments. Future research will explore multi-hop iterative retrieval strategies and the expansion of continuous domain-adaptive pre-training frameworks.

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